1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In BRCA2-deficient breast tumors, therapy_response to PARP inhibition (e.g., Olaparib) is determined by the pre-treatment intratumoral immune context: high baseline TIL density and elevated CD8A-expressing cytotoxic T cells enable PARP inhibitor–induced DNA damage to generate neoantigens that are effectively presented (via HLA-DQB1) and targeted by CD8+ T cells, producing durable anti-tumor immunity. BRCA2-deficient tumors with low TIL density or low CD8A expression will exhibit poor therapy_response despite genomic susceptibility. PARP1 serves as a secondary anchor that modulates the magnitude of DNA damage signaling, potentially amplifying or dampening the immune readout. Tier 2 modifiers (HLA-DQB1, CSF1, IL6, CRP, osteopontin) influence antigen presentation and the immunosuppressive microenvironment, further shaping the likelihood and durability of therapy_response. The testable predictions are: (1) among BRCA2-deficient breast cancers, those with high TIL density and high CD8A will show higher therapy_response to Olaparib; (2) modulators of antigen presentation and macrophage polarization (e.g., HLA-DQB1, CSF1) and systemic inflammation markers (IL6, CRP) will stratify responders from non-responders within this BRCA2-deficient cohort.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	BRCA2, CD8A
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

BRCA2: BRCA2 (DNA repair deficiency) and CD8A (cytotoxic T cell marker) because they mechanistically bridge genomic context to immune effector presence.

CD8A: CD8A to satisfy the 2-anchor-gene maximum and relegated PARP1 and other supporting factors to secondary_modifiers.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PARP1	Supporting modifier from knowledge graph

Secondary	HLA-DQB1	Supporting modifier from knowledge graph
Secondary	CSF1	Supporting modifier from knowledge graph
Secondary	OSTEOPONTIN	Supporting modifier from knowledge graph
Secondary	IL6	Supporting modifier from knowledge graph
Secondary	CRP	Supporting modifier from knowledge graph
Secondary	spermine	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective multi-omics analysis in BRCA2-deficient breast cancer cohorts with documented PARP inhibitor response (assessing TIL density, CD8A, HLA-DQB1, CSF1, IL6, CRP);

Prediction 2: in vitro co-culture systems of BRCA2-deficient breast cancer cells with CD8+ T cells ± PARP inhibitor to measure antigen presentation and cytotoxicity;

Prediction 3: in vivo BRCA2-deficient mouse models to test PARP inhibition with and without modulation of CSF1/IL6 pathways and to evaluate TIL recruitment and therapy_response. This approach provides a mechanistic, testable framework with clear clinical relevance for selecting BRCA2-mutant patients for PARP inhibitor therapy.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PARP1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HLA-DQB1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.